

Henry Tran

(617) 817-1860 | henrytran07@ucla.edu | henrytran07.github.io | github.com/henrytran07

EDUCATION

University of California, Los Angeles

Bachelor of Science in Electrical Engineering

Sep. 2026 – Expected June 2028

Los Angeles, CA

Mt. San Antonio College

Associate of Arts in Mathematics

Aug. 2022 – June 2026

Walnut, CA

- **Relevant Coursework:** Object-Oriented Programming in C++, Data Structures and Algorithms, Linear Algebra, Differential Equations, Electricity and Magnetism, Modern Physics, and Quantum Physics.

EXPERIENCE

Machine Learning Researcher

Algoverse AI Research

May 2025 – July 2026

Remote

- Co-authored *Before You Scale: The Cost of Supervision Mismatch in PRM Distillation*, accepted to the **COLM 2026 ER Workshop**.
- Traced distilled PRM failure to label rendering convention rather than student capacity; re-rendering generated teacher labels into the target first-error format raised ProcessBench-MATH ROC-AUC from **0.62 to 0.74**, and the resulting **0.3 GPU-hour** diagnostic replaces a ~ 15 GPU-hour recipe sweep

Team Lead Programmer

Mt. San Antonio College (Mt. SAC) Robotics Team

Aug. 2025 – June 2026

Walnut, CA

- Led an 8-person programming subteam to **13th** in Skills worldwide at the VEX U World Championship 2026 and the **highest** autonomous skills score in California.
- Built an autonomous navigation system in C++ with two-wheel odometry, IMU heading fusion, and PID controllers with anti-windup and derivative filtering, using the VEX Vision Sensor for goal alignment, earning the **Think Award** at EZ-Competition.
- Built a team website archiving the Push Back season, documenting CAD models, engineering notebooks, and source code.

PROJECTS

$\Sigma 4$ Electronic Guitar

Aug. 2026

- Built a full-octave electronic guitar in **48 hours** at UCLA HAcK: CircuitPython firmware on an **RP2350** driving a synthio engine over **I2S**, with a matrix keypad, pitch and volume pots, and 4 waveforms.
- Streamed JSON telemetry over USB serial into a **FastAPI** backend, driving a live browser visualizer with a canvas oscilloscope, note and frequency readout, and volume meter.

Fashion-MNIST Classifier from Scratch

July 2026

- Implemented a fully-connected neural network without an autograd framework, hand-deriving forward and backward passes with matrix calculus, including softmax/cross-entropy and batch normalization gradients
- Built He initialization, mini-batch gradient descent, and Adam optimization on a vectorized (784, m) data layout, reaching **90%** test accuracy across 10 clothing classes in 20 epochs.

Chess Engine

Summer 2025

- Designed an object-oriented chess engine from scratch with an abstract base class and polymorphic piece subclasses, each encapsulating its own legal-move logic.
- Implemented full rules enforcement (check/checkmate detection, promotion, and draw conditions), modeling board state with `std::set/std::map` and a custom position hash for efficient updates.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML, CSS

ML / Data: PyTorch, TensorFlow, Keras, Hugging Face Transformers, vLLM, scikit-learn, pandas, NumPy

Developer Tools: Git, GitHub, Jupyter Notebook, Visual Studio Code, Arduino IDE, VEX V5 SDK